

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

November – December 2011

CL 205 Building Materials and Construction

Date: 03.12.2011, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Give the detailed classification of engineering materials. Write in brief about any two alloys. [5]
- (b) Enlist the materials used as floor covering. Briefly explain brick flooring. [2]
- Q - 2 (a) Discuss about properties of fresh and hardened concrete. [5]
- (b) Differentiate thermo plastics and thermosetting plastics. [5]
- (c) What are the uses of ceramics and rubber? [4]

OR

- Q - 2 (a) Explain stepwise manufacturing process of bricks. [5]
- (b) Enumerate and discuss briefly types of industrial timber. [5]
- (c) Define Mortar. Write properties and uses of aggregate. [4]
- Q - 3 (a) Sketch various types of steps. Write dimensions of a step for residential building and public building. [4]
- (b) Define: Purlin, Rafter, Flight and Baluster. [4]
- (c) Write short notes on any Two types of roof covering. [3]
- (d) Discuss various factors affecting selection of flooring materials. [3]

OR

- Q - 3 (a) What are the requirements of a stair? Give the classification of stairs. Draw a neat sketch of bifurcated staircase. [6]
- (b) Draw a neat sketch of a King post truss having a span of 8.0 m showing all the components. [5]
- (c) Write a short note on suspended type of timber flooring. [3]

SECTION - II

- Q - 4 (a) Draw neat sketches of square rubble uncoursed and ashlar chamfered stone masonry. [2]
- (b) What are the requirements of good plaster? Draw sketches of various types pointing. [5]

- Q - 5 (a)** Design strip footing for a brick wall 30 cm thick and 3.3 m high above the ground level. The wall carries a superimposed load of 108 kN/m run. The soil has unit weight of 17 kN/m³, angle of repose is 30° and safe S.B.C. is 160 kN/m². The footing may have lime concrete base with unit weight of 20 kN/m³ and modulus of rupture equal to 154 kN/m². Assume unit weight of masonry as 19.5 kN/m³. [7]
- (b)** Draw a plan of alternate courses of English bond for 2 brick thick wall. [5]
- (c)** Explain briefly Tee-Junctions in brick work. [2]

OR

- Q - 5 (a)** Write short notes on following site exploration methods: [7]
(i) Wash boring (ii) Geo-physical methods
- (b)** Draw a plan of alternate courses of Single Flemish bond for 1½ brick thick wall. [5]
- (c)** Enumerate various types of composite masonry. Discuss any one in detail. [2]
- Q - 6 (a)** Draw neat sketch of battened, ledged, braced and framed door showing all components with horizontal and vertical cross sections. [6]
- (b)** Write a short note on pivoted windows. [2]
- (c)** Write short notes on following: [6]
(i) Auger boring (ii) Single flying shore (iii) Pit method of underpinning

OR

- Q - 6 (a)** Draw neat sketches of following (Any Two): [5]
(i) Casement window (ii) Fanlight ventilator (iii) Dormer and Gable window
- (b)** Enlist types of windows. Give the recommended dimensions of windows as per IS. [3]
- (c)** Write short note on: [6]
(i) Rotary boring (ii) Stay bracing (iii) Needle scaffolding
